

REGION-WISE ANALYSIS DASHBOARD



◆ PROJECT OVERVIEW

- This Tableau project provides a comprehensive **region-wise analysis of Sample Superstore data**, focusing on how **orders, sales performance and returns** vary across different geographic regions that highlights regional patterns, identify areas of strong and weak performance and understand the impact of product returns on overall business outcomes which further helps uncovering key trends and potential areas for improvement.
- The Deep Dive analysis moves beyond the overall regional KPIs to identify the specific **sub-categories, months and products** that evaluates multiple business metrics, including **Sales, Profit, Quantity, Discount, Order Count and Return Count**. The analysis reveals that the strongest performers are not necessarily consistent across all metrics. **A sub-category may generate the highest sales while another generates the highest profit**, highlighting the importance of evaluating performance from multiple business perspectives.
- A **key objective of the Deep Dive is to uncover relationships that may not be visible from overall KPIs alone**. For example, a **product may generate high sales but comparatively low or negative profit, while another product may generate fewer sales but contribute significantly to profitability**.



◆ THE PROBLEM STATEMENT:

- The primary objective of this project was to provide stakeholders with a **comprehensive region-wise view of Superstore performance** and transform transactional data into meaningful business insights. The analysis focuses on comparing key measures such as **sales, profit, quantity, discount, order count, product performance and returns** across the Central, East, South and West regions.
- Beyond high-level regional comparisons, stakeholders also require the ability to investigate **sub-category, product and monthly performance** to identify the strongest and weakest contributors within each region. The analysis therefore incorporates a **Top 5 deep-dive approach** to uncover high-performing products and sub-categories, recognize emerging trends and understand the relationship between sales, profitability, discounts, order volume, and returns.

- The ultimate goal is to support **data-driven strategic decision-making** for future Superstore initiatives and potential model expansion by using current business outcomes to identify opportunities, risks and performance patterns that can inform future planning and forecasting.



◆ DATASET DETAILS

- The data for this project was provided as part of the Introduction to **Business Analytics and Big Data coursework at Florida Atlantic University (FAU)**.
- The dataset consists of multiple interconnected sheets, including Orders, Region, Products, Customers, Returns and People. For this analysis, I focused specifically on the Orders and Returns tables because they provided the core information required to evaluate regional business performance and return activity.
- The **Orders table contains 209,895 records** and includes key attributes such as order and shipping dates, ship mode, customer information, geographic details, region, product category and sub-category, sales, quantity, discount and profit. **The Returns table contains 594 records** and provides return status information at the Order ID level.



◆ WHAT'S INSIDE

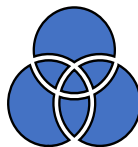
- **Data Preparation & Modeling** - Prepared and structured the Orders and Returns data for integrated analysis.
- **KPI Analysis** - Evaluated total sales, average sales, customers, orders, products, quantity, profit, discounts and returns.
- **Region-wise Comparison** - Compared Central, East, South, and West regions using interactive primary and secondary region selectors.
- **Geographic Analysis** - Visualized sales distribution and enabled users to explore performance across regions, states and cities.
- **Sub-Category Analysis** - Examined sales and profit performance across product sub-categories.
- **Trend Analysis**- Analyzed sales performance across quarters and months to identify changes and patterns over time.
- **Top 5 Analysis** - Identified leading sub-categories and products based on sales, profit, quantity, discount, orders, and returns.

- **Interactive Parameters** - Used dynamic region and metric selectors to allow users to control the analysis.
- **Dashboard Actions & Filters** - Connected visualizations through interactive filtering for a seamless analytical experience.
- **Deep Dive Analysis** - Enabled users to move from regional-level insights into state, city, product, sub-category, and monthly performance.
- **Tableau Storytelling** - Combined the overall regional analysis and detailed deep dive into a structured business analytics story.
- **Strategic Insights** - Designed the analysis to help stakeholders identify performance gaps, opportunities, trends, and areas that may support future planning and decision-making.



◆ DAX FORMULAS & NEW MEASURES

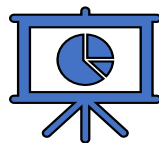
- Parameters were incorporated into the Tableau solution to make the dashboard interactive, flexible and user-driven. **The Primary Region and Secondary Region parameters allow users to select two regions and perform side-by-side comparisons** without creating separate dashboards for every regional combination.
- A **Region Selector** parameter was also introduced to distinguish between primary and secondary selections, while a **Select Metric** parameter enables users to switch the analytical focus among key measures such as Sales, Profit, Discount, and Quantity.
- Calculated fields were then created to connect the parameter selections with the corresponding worksheets. For example, the **Primary region calculation evaluates the selected value against the region parameter**, ensuring that the appropriate regional data is displayed dynamically. A similar calculation was created for the secondary region. Tableau sets were created for Top 5 Sub-Categories by Sales and Top 5 Sub-Categories by Profit.



◆ DATA MODELING

- I loaded the Sample Superstore data into Excel and reviewing the available fields for consistency and usability. Duplicate or unnecessary columns were removed and the **Orders and Returns datasets were combined through a union-based structure** to support integrated analysis of order performance and return activity.

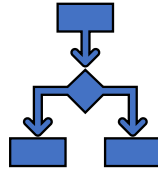
- Individual Tableau worksheets were created for the major business metrics, including **Total Sales, Average Sales, Customer Count, Total Orders, Total Products, Geographic Map, Sub-Category Analysis, Line Charts and Sub-Category Bar Charts**. Appropriate aggregations such as **SUM, AVERAGE and COUNT DISTINCT** were applied based on the analytical requirement of each measure.
- To enable dynamic regional comparisons, duplicate versions of the required worksheets were created. Two region parameters, **Primary Region** and **Secondary Region**, were introduced with selectable values for Central, East, South and West. The parameters were connected to the corresponding primary and comparison worksheets, allowing users to dynamically compare two regions within the same analytical view.
- The map uses **Sales to control the size of geographic markers**, while sub-category analysis incorporates aggregated measures such as maximum, minimum and sales values. A quarterly sales trend was also developed to evaluate how performance changes over time across regions.
- For deeper analysis, **Top 5 sets** were created for sub-categories based on total sales and total profit. Similar **Top 5 analyses were developed at the product and monthly levels, incorporating sales, profit, quantity, discount, order count and return count**. These structures allowed the dashboard to move beyond overall performance and identify the specific contributors driving regional results.



◆ VISUALS UTILIZED TO ANALYZE THE DATA

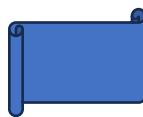
- The dashboard was designed using a combination of **KPI cards, geographic maps, tables, line charts, bar charts, and stacked charts** to provide multiple perspectives of the business data.
- The primary KPI section presents **Total Sales, Average Sales, Number of Customers, Total Orders, and Total Products** using text-based cards, allowing stakeholders to quickly understand the overall business position. A **light-themed geographic map of the United States** was incorporated to visualize the distribution of sales across locations, while a text table provides a detailed view of product **sub-category performance**.
- To analyze trends over time, a **line chart** was used to compare regional sales across different quarters. A **sub-category bar chart** provides a direct comparison of sales performance across product sub-categories, with unnecessary field headers hidden to maintain a cleaner visual presentation.
- For the deep-dive analysis, **stacked bar charts** were used to examine the Top 5 sub-categories based on sales and profit. **Stacked column charts** were used to analyze Top 5 performance by product name and month, allowing users to investigate the contributors behind regional

results. All visualizations were integrated into the **Region-wise Analysis dashboard** and connected to the Primary and Secondary Region parameters.



◆ STEPS I FOLLOWED TO BUILD THE DASHBOARD

- The primary dashboard, titled **“Region-wise Analysis,”** was designed as an interactive analytical view that brings together the major **performance metrics, visualizations, and regional comparison controls**. Primary and duplicate worksheets were placed within the dashboard, with the Primary Region parameter controlling the primary analysis and secondary is for duplicate sheets.
- To create a seamless interactive experience, **dashboard-level filter actions were implemented**. A Primary Filter was configured to connect the primary worksheets, while a Secondary Filter was configured for the corresponding comparison worksheets.
- Interactive dashboard actions were then applied to connect the deep-dive visualizations. Users can select a region from the map and drill further into a specific state or city, allowing the analysis to progress from regional performance to more granular geographic insights. This creates an analytical flow from Region → City → Sub-Category, making the dashboard useful for both executive-level review and detailed investigation.



◆ THE STORYTELLING

- The final Tableau story combines the project's **Region-wise Analysis** and **Deep Dive Analysis** into a structured analytical journey. The Region-wise Analysis provides stakeholders with a comprehensive overview of performance across the Central, East, South, and West regions, highlighting key metrics, geographic patterns, trends, and comparative performance.
- The Deep Dive Analysis then moves beyond overall regional results to identify the **Top 5 performers and key contributors** within each region. By examining products, sub-categories, months, sales, profit, quantity, discounts, orders, and returns, the analysis helps uncover areas of strength, potential improvement, and performance risks.



◆ WHY THIS PROJECT MATTERS

- This project demonstrates how raw transactional data can be **transformed into an interactive business intelligence solution that supports real-world decision-making**. Rather than presenting sales and return information as isolated numbers,
- The regional comparison **functionality allows stakeholders to quickly identify high- and low-performing regions**, while the **deep-dive analysis helps uncover the specific products, sub-categories and time periods contributing to those results**. This makes it possible to move from a high-level question such as **“Which region is performing better?”** to **more actionable questions such as “What products are driving that performance?” or “Where are potential performance or return issues occurring?”**. From a business perspective, these insights can support performance monitoring, product strategy, regional planning, opportunity identification, and future business expansion.



◆ KEY INSIGHTS

❖ Overall Regional Performance

- Among all regions, the **West region emerged as the strongest overall performer, generating the highest total sales of \$725,458, the highest total profit of \$108,418, the highest number of orders at 3,203, and the highest quantity sold at 12,266. It also served the highest number of customers at 686**. These results indicate that the West region has the strongest overall business contribution across both revenue generation and operational volume.
- In contrast, the **South region recorded the lowest overall sales, with total sales of \$391,722, along with the lowest number of customers (512), products (1,057), orders (1,620) and quantity sold (6,209)**. This suggests that the South region has comparatively lower business activity and may represent an opportunity for further investigation and targeted growth strategies.
- The difference between the **highest-performing West region and the lowest-performing South region is substantial, with the West generating approximately 85% more sales than the South**. This regional performance gap highlights the importance of understanding the factors contributing to stronger sales, customer activity, product demand and profitability in the West.

❖ Highest-Performing Sub-Category

- Across the regional product analysis, **Chairs in the West region generated the highest sales at \$101,781**, making it the strongest individual sub-category sales result provided in the analysis.
- The West-region Chairs performance included a **maximum sales value of \$2,804 a minimum value of \$41 and a discount value of 41**. The strong sales contribution from Chairs suggests that this product category is an important revenue driver for the West region and could be examined further for customer demand, pricing, discount strategy and future inventory planning.
- The **Phones sub-category in the East region was another exceptionally strong performer, generating \$100,615 in sales, only \$1,166 below West-region Chairs. Phones recorded a maximum value of \$4,549, a minimum of \$3, and a discount value of 44**. This indicates that technology-related products are particularly significant contributors to East-region sales.
- Interestingly, **Phones were also the highest-performing sub-category in the South region, generating \$58,304 in sales, with a maximum of \$4,368, minimum of \$8, and discount of 14**. Although the South's Phone sales were considerably lower than the East's, the repeated appearance of Phones as a top performer suggests that this category has meaningful demand across multiple regions.

❖ Lowest-Performing Sub-Category

- **Fasteners consistently appeared as the lowest-sales sub-category across all four regions.** This provides one of the clearest patterns in the analysis.
 - Central: \$778 sales - Max: \$58, Min: \$2, Discount: \$7
 - East: \$820 sales - Max: \$41, Min: \$1, Discount: \$6
 - South: \$503 sales - Max: \$55, Min: \$2, Discount: \$2
 - West: \$923 sales - Max: 93, Min: \$2, Discount: \$2
- Among these, the South region recorded the **lowest Fasteners sales at only \$503, while the West recorded the highest Fasteners sales at \$923**. Even though West has the strongest overall performance, Fasteners remain a relatively low-sales category compared with its other products.
- The consistency of Fasteners appearing at the bottom across all regions may indicate comparatively lower demand, limited market potential, product positioning issues, or an opportunity to review pricing, promotion and inventory strategies.

❖ Regional KPI Comparison

- The **Central** region generated \$501,240 in sales, with \$39,706 profit, 629 customers, 1,310 products, 2,323 orders, and 8,780 units sold. Its highest-performing category was Chairs at \$85,231, while Fasteners represented the lowest-performing category at \$778.
- The **East** region generated \$678,781 in sales and \$91,523 profit, supported by 674 customers, 1,422 products, 2,848 orders, and 10,618 units sold. Phones were the leading category at \$100,615, while Fasteners generated only \$820. The East therefore demonstrates strong revenue and profitability, particularly through its technology-related product performance.
- The **South** region recorded the lowest total sales at \$391,722, with \$46,749 profit, 512 customers, 1,057 products, 1,620 orders, and 6,209 units sold. Phones were its strongest category at \$58,304, while Fasteners had the lowest sales at \$503. The comparatively lower activity across almost every KPI makes the South an important region for further investigation.
- The **West** region was the strongest overall performer, achieving \$725,458 in sales and \$108,418 profit, supported by 686 customers, 1,509 products, 3,203 orders, and 12,266 units sold. Chairs were the strongest category at \$101,781, while Fasteners remained the lowest at \$923.

❖ Profitability Insight

- The regional results show that the West region not only leads in sales but also generates the highest profit, suggesting that its strong performance is not driven solely by sales volume. Based on the provided figures, West has an approximate profit-to-sales ratio of **14.9%**, compared with approximately 13.5% for East, 11.9% for South and 7.9% for Central.
- This indicates that the West appears to have the strongest overall profitability efficiency among the four regions. Central, despite generating more than **\$500K in sales**, produces comparatively lower profit, making its discounting, product mix and cost structure areas worth investigating.

❖ Customer and Order Activity

- The West region also leads in customers, orders and quantity sold, reinforcing its position as the most active region. Its **3,203 orders and 12,266 units sold** indicate a strong level of transactional activity.
- The East follows closely, with 2,848 orders and 10,618 units sold, while Central records 2,323 orders and 8,780 units. South has the lowest activity with 1,620 orders and 6,209 units.
- This pattern demonstrates that regional sales differences are associated with differences in customer reach, order volume and quantity sold, rather than being explained by a single metric.

❖ Average Sales Insight

- The South region has the highest average sales value at \$242, followed by East at \$238, West at \$226, and Central at \$216.
- This is an important insight because the South has the lowest total sales but the highest average sales value. Therefore, its weaker overall performance does not necessarily mean that individual transactions are less valuable. Instead, the lower number of customers and orders appears to be a more significant contributor to its overall sales gap.

❖ Discount Insight

- Discount patterns also vary considerably across products and regions. For example, **East-region Phones have a discount value of 44, while West-region Chairs have a discount value of 41. At the lower end, South and West Fasteners have discounts of only 2.**
- The Central region's highest-performing Chairs have a discount of 30, while its lowest-performing Fasteners have a discount of 7. These differences provide an opportunity for stakeholders to investigate whether discounting is contributing to higher sales or potentially affecting profitability.

❖ Key Business Takeaways

- Overall, the analysis highlights several important patterns:
 1. West is the strongest overall region, leading in sales, profit, customers, orders, and quantity sold.
 2. South is the weakest region by overall sales and activity, making it a potential area for targeted growth and customer acquisition strategies.
 3. West Chairs (\$101,781) represent the highest product/sub-category sales result provided, closely followed by East Phones (\$100,615).
 4. Fasteners consistently rank as the lowest-sales sub-category across all four regions, with the lowest result occurring in South at \$503.
 5. Phones demonstrate strong demand, appearing as the highest-performing category in both East and South.
 6. South has the highest average sales (\$242) despite having the lowest total sales, indicating that its major challenge may be transaction and customer volume rather than transaction value.
 7. West demonstrates the strongest profitability efficiency, with approximately 14.9% profit relative to sales, while Central has the lowest at approximately 7.9%.

8. The significant differences in discounts, sales, profit, and product performance suggest that pricing and discount strategies should be evaluated alongside product demand.
9. The regional performance gap indicates that stakeholders can use the West as a benchmark for identifying successful business patterns, while the South can be analyzed to identify opportunities for improvement.
10. The combination of regional, product, customer, order, and time-based analysis provides a foundation for strategic planning, resource allocation, product strategy, and future performance forecasting.

❖ DEEP DIVE ANALYSIS

➤ Sales Performance

- Among the four regions, the **West region recorded the highest total sales at \$725,458**, followed by East at \$678,781, Central at \$501,240, and South at \$391,722.
- At the deep-dive sub-category level, the highest sales contribution was:
 - **West — Chairs: \$101,781**
 - **East — Chairs: \$100,615**
 - **Central — Chairs: \$85,231**
 - **South — Phones: \$58,304**

Therefore, **West Chairs generated the highest sub-category sales value among the four regions at \$101,781.**

The lowest sub-category sales values provided were:

- **West — Binders: \$55**
- **South — Tables: \$35,768**
- **East — Tables: \$39,140**
- **Central — Tables: \$39,155**

The extremely low West Binders value of \$55 makes it the lowest sales result in the Deep Dive data provided.

❖ Profit Performance

- At the overall regional level, **West generated the highest total profit at \$108,418**, followed by East at \$91,523, South at \$46,749, and Central at \$39,706.

At the sub-category level:

- **West — Copiers: \$19,327**

- **East — Copiers: \$17,023**
- **Central — Copiers: \$15,609**
- **South — Phones: \$10,767**

Therefore, **West Copiers generated the highest sub-category profit at \$19,327** among the values provided.

The lowest sub-category profit was:

- **Central — Binders: -\$1,044**
- **South — Copiers: \$3,659**
- **East — Paper: \$9,015**
- **West — Phones: \$9,111,961 as provided**

Central Binders therefore represent the clearest negative-profit result, with a loss of **\$1,044**.

❖ **Highest Sales by Sub-Category**

- The strongest sub-category sales performance across the regions was recorded by **Chairs in the West region, generating \$101,781**. East Chairs followed closely at \$100,615, while Central Chairs generated \$85,231.
- South differed from the other three regions, with **Phones generating the highest sub-category sales at \$58,304**.
- This indicates that **Chairs are a consistently strong sales driver across Central, East, and West**, while Phones play a particularly important role in South.

❖ **Lowest Sales by Sub-Category**

- The lowest result in the provided Deep Dive data is **West Binders at \$55**. This is significantly lower than the other lowest-performing sub-category values and should be investigated further to determine whether it reflects genuinely low demand, limited orders, or a data/filtering issue.
- For Central, East, and South, **Tables** were the lowest-performing sub-category among the values provided:
 - Central Tables - \$39,155
 - East Tables - 39,140
 - South Tables - \$35,768

❖ **Highest Profit by Sub-Category**

- **West Copiers generated the highest sub-category profit at \$19,327**, making Copiers a particularly important profitability driver in the West.
- Copiers also ranked highest in Central (\$15,609) and East (\$17,023), demonstrating a consistent profitability pattern. South differs again, with **Phones generating the highest profit at \$10,767**.

❖ **Lowest Profit by Sub-Category**

- The lowest profit result was **Central Binders at -\$1,044**, meaning the sub-category generated a loss rather than a positive contribution.
- South Copiers generated \$3,659, while East Paper generated \$9,015. These differences demonstrate that the product category generating high sales does not necessarily generate the highest profit.

❖ **Business Interpretation**

- The sub-category analysis demonstrates an important distinction between **revenue leadership and profitability leadership**. For example, Chairs dominate sales in several regions, while Copiers dominate profit in Central, East, and West. This suggests that stakeholders should evaluate both sales and profit before making decisions regarding inventory, promotions, and product expansion.

❖ **Sales by Month**

- Monthly sales show noticeable seasonal variation across the four regions.
 - **Central:** September — \$76,833
 - **East:** November — \$69,364
 - **South:** November — \$59,539
 - **West:** December — \$77,928
- Therefore, **West December recorded the highest monthly sales value at \$77,928** among the regional monthly results provided.

The lowest monthly sales values were:

- Central — March: \$41,216
- East — October: \$23,749
- South — October: \$24,202
- West — March: \$31,949

➤ The **East region in October** recorded the lowest monthly sales at **\$23,749**.

❖ Profit by Month

➤ The strongest monthly profit results were:

- Central — October: \$12,748
- East — September: \$7,479
- South — March: \$6,988
- West — November: \$5,524

➤ Therefore, **Central October** recorded the highest monthly profit at **\$12,748** among the values provided.

The lowest monthly profit values were:

- Central — March: \$1,226
- East — October: \$1,242
- South — October: **-\$1,676**
- West — March: **-\$3,586**

The **West region in March** recorded the lowest monthly profit at **-\$3,586**, indicating a negative profitability period.

❖ Quantity by Month

The highest quantity values provided were:

- Central — November: 1,422
- East — November: 733
- South — November: 90*
- West — December: 880

Therefore, **Central November** recorded the highest monthly quantity at **1,422** among the supplied figures.

The lowest values were:

- Central — March: 656
- East — October: 326
- South — March: 459*

- West — March: 328

East October recorded the lowest supplied quantity at **326**.

❖ **Discount by Month**

The highest monthly discount values were:

- Central — November: 84
- East — September: 44
- South — November: 40
- West — December: 44

Therefore, **Central November had the highest discount value at 84**.

The lowest values were:

- Central — March: 45
- East — October: 15
- South — March: 19
- West — March: 14

Therefore, **West March recorded the lowest discount value at 14**.

❖ **Orders by Month**

The highest monthly order counts were:

- Central — November: 371
- East — November: 191
- South — November: 227
- West — December: 223

Therefore, **Central November had the highest order count at 371**.

The lowest monthly order counts were:

- Central — March: 177
- East — October: 88
- South — March: 124
- West — March: 89

Therefore, **East October** recorded the lowest order count at **88**.

❖ Returns by Month

The highest return counts were:

- Central — December: 8
- East — September: 5
- South — December: 5
- West — December: 23

Therefore, **West December** recorded the highest number of returns at **23**.

The lowest return count was **1**, occurring in South during October and March. East also recorded 2 returns in June, August, and November.

❖ Seasonal Pattern

- A notable pattern emerges from the monthly analysis: **November and December frequently appear among the strongest months for sales, quantity and order activity**, particularly in East, South and West. In contrast, **March and October frequently appear among the weaker months**, especially for sales, profit and order activity.

❖ Central Region

Sales: Canon image was the strongest product at **\$17,500**, while GBC was the lowest at **\$2,691**.

Profit: Canon image generated the highest profit at **\$8,400**, while Fellowes recorded the lowest profit at **-\$1,525**, indicating a loss.

Quantity: HON 4500 was reported as the highest at **12**, while Fellowes was reported as the lowest at **12**. These two values appear inconsistent in the supplied data and should be verified in Tableau.

Discount: Fellowes had the highest discount at **2**, while Canon image had the lowest at **0**.

Orders: Fellowes was reported at **3 orders**, while Canon had the lowest at **1 order**.

Overall, Canon image is the strongest Central-region product from a **sales and profit perspective**, while Fellowes requires attention because of its negative profit.

❖ East Region

Sales: Canon image led with **\$30,100**, while Fellowes was lowest at **\$5,592**.

Profit: Canon image generated the highest profit at **\$10,080**, while HON recorded the lowest at **-\$210**.

Quantity: HON recorded the highest quantity at **15**, while Fellowes was lowest at **5**.

Discount: GBC and Canon had the highest discount at **1**, while HON and Fellowes recorded **0**.

Orders: GBC had the highest order count at **4**, while HON and Fellowes had the lowest at **2**.

Canon image is therefore the strongest East-region product based on both **sales and profit**, while HON is a product requiring further profitability investigation.

❖ South Region

Sales: Cisco generated the highest sales at **\$22,638**, while HON recorded the lowest at **\$2,804**.

Profit: Fellowes generated the highest profit at **\$3,813**, while GBC recorded the lowest at **-\$1,811**.

Quantity: GBC was reported as highest at **17**, while HON was lowest at **5**.

Discount: GBC and Cisco had the highest discount at **1**, while HON and Fellowes had **0**.

Orders: GBC recorded the highest order count at **5**, while HON and Cisco recorded the lowest at **1**.

South demonstrates an interesting difference between revenue and profitability: **Cisco leads sales, but Fellowes leads profit**. This reinforces the importance of evaluating multiple metrics rather than relying solely on sales.

❖ West Region

Sales: The supplied values state Canon image at **\$1,400** and HON at **\$3,925**. Based on these numbers, **HON should be the highest and Canon image should be the lowest**, rather than the labels currently provided.

Profit: Canon image generated the highest profit at **\$6,720**, while HON recorded **\$0**.

Quantity: Fellowes had the highest quantity at **8**, while Canon had the lowest at **4**.

Discount: Fellowes had the highest discount at **1**, while Canon had the lowest at **0**.

Orders: Fellowes had the highest order count at **3**, while Canon had the lowest at **1**.

The West product-level results therefore show that **Canon image has strong profitability despite the supplied sales figure being lower than HON**, while Fellowes leads in quantity and order activity.

❖ Cross-Region Product Insight

Canon image is the most consistently strong product in the Deep Dive analysis, particularly in Central and East, where it leads both sales and profit. It also leads profit in West.

The analysis also identifies several products with **negative profit**, including Central Fellowes (-\$1,525), East HON (-\$210), and South GBC (-\$1,811). These products may require further investigation into pricing, discounts, product costs, or order economics.

The product-level results demonstrate that **high sales, high quantity, high order count, and high profit do not always belong to the same product**. This is an important business insight because product strategy should consider the complete performance picture rather than a single KPI.

❖ The Deep Dive analysis identifies several key business patterns across the four regions.

1. West leads overall performance.

West has the highest overall sales (**\$725,458**) and profit (**\$108,418**). Its strongest sub-category sales come from **Chairs (\$101,781)**, while **Copiers (\$19,327)** generate the highest sub-category profit.

2. Chairs are a major sales driver.

Chairs are the highest-sales sub-category in Central, East, and West. West Chairs represent the strongest sub-category sales result in the analysis at **\$101,781**.

3. Phones are particularly important in South.

Phones generate the highest sales (**\$58,304**) and profit (**\$10,767**) among South's sub-categories.

4. Copiers are consistently strong for profitability.

Copiers are the highest-profit sub-category in Central, East, and West, demonstrating a recurring profitability pattern across regions.

5. Binders require attention in Central.

Central Binders generated **-\$1,044 profit**, making them the lowest-profit sub-category among the supplied values.

6. West Binders show an unusually low sales value.

West Binders are reported at only **\$55 in sales**, which is dramatically lower than the other regional lowest-performing sub-categories. This should be validated against the Tableau source data.

7. November and December demonstrate strong activity.

November is the strongest month for sales in East and South, while December leads West sales. November also records the highest quantity and order activity in several regions.

8. March is generally a weak month.

March repeatedly appears as a low-performing period for sales, quantity, orders, and profit. West March is particularly concerning because its profit falls to **-\$3,586**.

9. West December has the highest return count.

West December recorded **23 returns**, the highest monthly return count in the supplied data. This should be investigated alongside December's strong sales to determine whether higher transaction volume is contributing to increased returns.

10. Canon image is a strong product performer.

Canon image leads sales in Central (**\$17,500**) and East (**\$30,100**) and leads profit in Central (**\$8,400**), East (**\$10,080**), and West (**\$6,720**).

➤ Overall, the Deep Dive provides a more granular understanding of the business by moving from **Region → Sub-Category → Month → Product**. This structure helps stakeholders identify not only where performance is strongest or weakest, but also **what products and time periods are responsible for those outcomes**. These findings can support future decisions around inventory planning, product strategy, pricing, promotions, regional resource Allocation and demand forecasting.

Final Deep Dive Highlights

- **Highest overall regional sales: West — \$725,458**
- **Lowest overall regional sales: South — \$391,722**
- **Highest overall regional profit: West — \$108,418**
- **Lowest overall regional profit: Central — \$39,706**

❖ Sub-Category

- **Highest sub-category sales: West — Chairs, \$101,781**
- **Lowest sub-category sales: West — Binders, \$55**
- **Highest sub-category profit: West — Copiers, \$19,327**
- **Lowest sub-category profit: Central — Binders, -\$1,044**

❖ Monthly

- Highest monthly sales: West — December, \$77,928
- Lowest monthly sales: East — October, \$23,749
- Highest monthly profit: Central — October, \$12,748
- Lowest monthly profit: West — March, -\$3,586
- Highest monthly quantity: Central — November, 1,422
- Lowest monthly quantity: East — October, 326
- Highest monthly discount: Central — November, 84
- Lowest monthly discount: West — March, 14
- Highest monthly orders: Central — November, 371
- Lowest monthly orders: East — October, 88
- Highest monthly returns: West — December, 23
- Lowest monthly returns: South — October and March, 1 each

❖ Product Name

- Highest product sales: East — Canon Image, \$30,100
- Lowest product sales: West — HON, \$3,925
- Highest product profit: East — Canon Image, \$10,080
- Lowest product profit: South — GBC, -\$1,811
- Highest product quantity: South — GBC, 17
- Lowest product quantity: Central — Canon Image, 5
- Highest product discount: Central — Fellowes, 2
- Lowest product discount: Central — Canon Image, 0
- Highest product orders: South — GBC, 5
- Lowest product orders: Central — Canon Image, 1

➤ One particularly strong insight is that **the same product does not necessarily lead every metric**. For example, **East Canon Image leads both sales and profit**, while **South GBC leads quantity and orders but generates negative profit of \$1,811**. That's exactly the type of

business insight that makes your Tableau Deep Dive more valuable than simply showing rankings.

- Another strong finding is the **West region's December performance**: it has the highest monthly sales (**\$77,928**) and the highest returns (**23**) at the same time. That creates an interesting business question whether the high returns are simply associated with increased transaction volume or whether there is a sub-category issue worth investigating.



❖ STRATEGIC RECOMMENDATIONS

1. Prioritize the West Region as a Benchmark for Growth

- The West region is the strongest overall performer, generating the highest total sales (**\$725,458**) and profit (**\$108,418**), while also leading in customers, orders and quantity sold. Management should study the factors contributing to West's performance and use them as a benchmark when developing strategies for other regions. Successful product combinations, customer segments, pricing approaches, promotional strategies, and operational practices from the West can be evaluated for potential application in lower-performing regions.

2. Develop a Targeted Growth Strategy for the South Region

- The South region records the lowest overall sales (**\$391,722**), customers (**512**), orders (**1,620**), and quantity sold (**6,209**). However, its average sales value is the highest among the four regions at **\$242**. This suggests that the primary opportunity may not be increasing transaction value, but increasing **customer reach and order frequency**. Management should consider targeted customer acquisition campaigns, regional promotions, retention initiatives, and market-specific product strategies to increase the number of transactions while maintaining the region's relatively strong average sales value.

3. Strengthen High-Performing Product Categories

- Chairs are the strongest sales-driving sub-category in Central, East, and West, with **West Chairs generating \$101,781**, the highest sub-category sales result in the analysis. Phones are particularly important in the South region, generating **\$58,304 in sales and \$10,767 in profit**. Management should ensure appropriate inventory availability for consistently high-performing categories and evaluate whether successful products can be expanded into regions where demand is currently lower.

4. Use Profitability, Not Sales Alone, for Product Decisions

- The analysis demonstrates that the product with the highest sales is not always the most profitable. For example, South Cisco generates the highest product sales at **\$22,638**, while Fellowes generates the highest product profit at **\$3,813**. Product decisions should therefore consider a combination of **sales, profit, quantity, discounts and order activity** rather than using sales as the only performance indicator. This can help prevent investment in high-revenue products that provide limited or negative profitability.

5. Investigate Negative-Profit Products

- Several product-level results show negative profitability, including **South GBC (-\$1,811)**, **Central Fellowes (-\$1,525)**, and **East HON (-\$210)**. These products should be reviewed to understand whether negative profit is associated with excessive discounts, product costs, pricing, low order volume, or other operational factors. A product-level profitability review could help determine whether these items require **repricing, promotional changes, supplier-cost optimization, inventory reduction, or product portfolio reconsideration**.

6. Review Discount Strategy

- Discount levels vary considerably across regions and products. Central November recorded the highest monthly discount value at **84**, while West March recorded the lowest at **14**. Rather than applying broad discounting strategies, management should evaluate the relationship between **discount percentage, incremental sales and resulting profit**. Discounts should ideally be targeted toward products or customer segments where they generate sufficient additional demand without unnecessarily reducing margins.

7. Prepare for Strong November–December Demand

- November and December frequently appear among the strongest periods for sales, quantity, and order activity. West recorded its highest monthly sales in December at **\$77,928**, while Central recorded its highest monthly quantity (**1,422**) and order count (**371**) in November. Management should use these patterns for **inventory planning, staffing, supply chain preparation, promotional scheduling, and customer-support capacity**. Preparing inventory and operational resources before these high-demand periods can reduce the risk of stock shortages and service issues.

8. Investigate the High Return Activity in West December

- West December recorded **23 returns**, the highest monthly return count in the Deep Dive analysis, while also generating the highest monthly sales at **\$77,928**. Because higher sales volume can naturally result in more returns, the relationship should be examined using a **return-rate metric**, rather than relying only on the absolute return count. Management should identify the products, categories, locations, or customer segments contributing most to the returns and determine/ whether the issue is volume-driven or indicates a product quality concern.

❖ Overall Strategic Direction

- The analysis suggests that the strongest strategy is **not simply to maximize sales**, but to optimize the combination of **revenue, profitability, customer growth, product mix, discount effectiveness, and return management**. The West region provides a useful benchmark for understanding successful performance, while the South presents the greatest opportunity for customer and order growth. At the product level, management should continue supporting high-performing categories such as **Chairs, Phones and Copiers**, while investigating negative-profit and persistently low-performing products.